MILIND B PRABHU

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EDUCATION

University of Michigan, Ann Arbor — *September 2022 - Present*
PhD in Theoretical Computer Science, advised by Nikhil Bansal — USA

Indian Institute of Technology, Guwahati — *July 2018 - July 2022*
Bachelor of Technology in Computer Science and Engineering — India

- GPA: 9.32/10.0

PUBLICATIONS

- **Coresets for Capacitated Clustering via Dual Concentration** — **SODA 2027**
  with Chris Schwiegelshohn, Sudarshan Shyam, and Erik Waingarten
- **The Cube-Root Phenomenon in Online Carpooling** — **ITCS 2027**
  with Nikhil Bansal, Sahil Singla, and Siddharth M. Sundaram
- **Fault Tolerant Coresets** — **NeurIPS 2026**
  with Chris Schwiegelshohn and Sudarshan Shyam
- **Online Graph Balancing and the Power of Two Choices** — **FOCS 2026**
  with Nikhil Bansal, Sahil Singla, and Siddharth M. Sundaram
- **Approximation Preserving Coresets** — **ICML 2026**
  with Chris Schwiegelshohn and Sudarshan Shyam
- **Sensitivity Sampling for k-Means: Worst Case and Stability Optimal Coreset Bounds** — **FOCS 2024**
  with Nikhil Bansal, Vincent Cohen-Addad, David Saulpic, and Chris Schwiegelshohn
- **Learning Multiple Secrets in Mastermind** — **ICML 2024**
  with David Woodruff
- **On Minimizing Generalized Makespan on Unrelated Machines** — **APPROX 2023**
  with Nikhil Ayyadevara and Nikhil Bansal
- **Generalizing Greenwald-Khanna Streaming Quantile Summaries for Weighted Inputs** — **ICDT 2023**
  with Sepehr Assadi, Nirmit Joshi, and Vihan Shah

PREPRINTS AND MANUSCRIPTS

- **Non-Clairvoyant Scheduling is Hard Even for Trees** — **Under submission**
  with Kunal Agrawal, Owen Druzgal, and Jinhao Zhao
- **Compress, Solve, and Lift: Coresets for Fair Division** — **Under submission**
  with Sudarshan Shyam
- **A (4m - 5)-Competitive Deterministic Algorithm for Online Transportation** — **Under submission**
  with Aditya Anand, Jonathan Conroy, Ian DeHaan, Gary Hoppenworth, and Chaitanya Nalam
- **The Price of Capacities for Clustering Coresets** — **Under submission**
  with Chris Schwiegelshohn and Sudarshan Shyam

TALKS

- **Microsoft Research India, Jan 2025.** Online talk on *Coresets for k-Means Clustering*.
- **Program in Algorithmic & Combinatorial Thinking 2024 (PACT).** Talk on randomized load balancing for high-school and undergraduate students.
- **New Horizons in TCS Summer School, 2021.** Virtual teaching assistant for a summer school introducing under-represented undergraduate students to theoretical computer science.
- **Program in Algorithmic & Combinatorial Thinking 2020 (PACT).** Talks on sublinear algorithms, streaming algorithms, and communication complexity.

SERVICE AND PROFESSIONAL ACTIVITIES

- **External reviews:** STOC 2025; SODA 2025; FSTTCS 2024.
- **Department service:** Co-organizer of the weekly *Theory Lunch* at the University of Michigan in Fall 2024 and Winter 2025.

WORKSHOP PARTICIPATIONS

- **Efficient Algorithms for High Dimensional Metrics: New Tools, 2024.** DIMACS, Rutgers University.
- **Old Questions and New Directions in Theory of Clustering, 2024.** EnCORE Institute, University of California, San Diego.
- **PACT 2019.** Program in Algorithmic & Combinatorial Thinking, Princeton University.

AWARDS AND HONORS

- **IIT-JEE Advanced 2018:** Ranked 582nd among one million candidates.
- **Indian National Mathematical Olympiad 2017:** Certificate of merit.
- **KVPY 2016:** Fellowship from the Department of Science and Technology, Government of India; all-India rank 162.
- **NTSE 2016:** National Talent Search Examination Scholarship.